Sensitivity Check: SCR → SCA Reclassification of the Two High Pu-239 Total-XS Points

Round-3 TNC fit (2026-06-23 database). Sensitivity test prompted by G. Noguère's query on the scattering classification of the two highest Pu-239 total-cross-section measurements.

Question

Two Pu-239 total-cross-section (transmission) measurements sit clearly above the rest of the Pu-239 (n,tot) set (which clusters at ~1022–1030 b):

No.	Author	Year	Quantity (round 3)	Value	σ
75	Pattenden	1956	(ABS 39)+SCR 39	1039 b	±25 b (2.41%)
192	Anderson	1945	(ABS 39)+SCR 39	1045 b	±25 b (2.39%)

Both currently carry their scattering component as **SCR** (the "rolled metallic sample" / reduced-coherent convention). The question raised is whether that assignment should instead be **SCA** (true elastic). This note refits the round-3 database with both entries changed to `(ABS 39)+SCA 39` and compares against the round-3 baseline. Nothing else is altered.

Result

Quantity	Round 3 (SCR)	Variant (SCA)	Δ	Δ%
χ^2	73.42	73.36	-0.07	—
SCA 39	7.4805	7.4805	-0.0001	-0.0008%
SCR 39	6.7373	6.6958	-0.0415	-0.6165%
ABS 39	1022.0765	1022.0701	-0.0063	-0.0006%
FIS 39	750.0765	750.0742	-0.0022	-0.0003%
CA 39	272.0000	271.9959	-0.0041	-0.0015%
ALPHA 39	0.3626	0.3626	-0.0000	-0.0012%
σ_tot 39 = ABS+SCA	1029.5570	1029.5506	-0.0064	-0.0006%
K1 = F3ETA 39	1172.5370	1172.5353	-0.0017	-0.0001%

Other nuclides (unaffected — shown for completeness):

Quantity	Round 3	Variant	Δ%
σ_tot 33	588.3303	588.3302	-0.0000%
σ_tot 35	698.5938	698.5932	-0.0001%
σ_tot 41	1392.6215	1392.6209	-0.0000%
K1 = F3ETA 33	742.0951	742.0951	-0.0000%
K1 = F3ETA 35	723.3823	723.3818	-0.0001%
K1 = F3ETA 41	1704.6467	1704.6464	-0.0000%

Normalized residual (pull) of the two reclassified points, before and after:

No.	Author	pull (SCR)	pull (SCA)
75	Pattenden	+0.41	+0.38
192	Anderson	+0.65	+0.62

Interpretation

The classification has no practical consequence for the evaluation. Every physically relevant quantity — ABS, FIS, CA, the total cross section σ_tot, and the Westcott-folded K1 = F3ETA — for Pu-239 and for all other nuclides is stable to **< 0.002** %, and χ^2 improves by a negligible 0.07. The only visibly moving parameter is **SCR(^{239}Pu)**, which shifts by ≈ −0.6 %; it is itself weakly determined (≈ 16 % uncertainty) and does not feed the recommended thermal constants.

The reason is quantitative: the fitted Pu-239 scattering terms differ by only SCA − SCR ≈ 7.48 − 6.74 = **0.74 b**. Re-labelling a point from ABS+SCR to ABS+SCA changes its predicted total by exactly this ~0.74 b, which is negligible against the **±25 b (2.4 %)** uncertainties of Pattenden and Anderson. Both points are also weak constraints on ABS(^{239}Pu) — which is pinned by many tighter entries (Bollinger, Safford, Palevsky, Spencer, Zimmerman at 0.5–1.3 %) — so moving 0.74 b of their value between the SCR and SCA channels barely perturbs the fit. Consistently, their pulls stay below 1σ in both cases (Pattenden +0.41⟶+0.38, Anderson +0.65⟶+0.62).

Conclusion. The SCR-vs-SCA assignment of Pattenden 1956 (No. 75) and Anderson 1945 (No. 192) is physically debatable but numerically immaterial at the round-3 level: the recommended Pu-239 constants and K1 are identical to within rounding whichever convention is adopted. There is therefore no need to resolve the question to finalise the fit; it can be settled on physical grounds (sample form / extinction corrections in the original experiments) without affecting the results.

Reproducibility

- Script: `evaluation/sensitivity_scr2sca_75_192.py`
- Variant database: `axton1986/axton1986_data_capote_round3_scr2sca_75_192.json` (= round-3 DB with Nos. 75 and 192 changed to `(ABS 39)+SCA 39`)
- Baseline: `axton1986/axton1986_data_capote_round3.json` and `evaluation/report_capote_round3.md`